Enterprise Knowledge Assistant

Project reference notes for future README editing, setup, evaluation, and interview preparation.

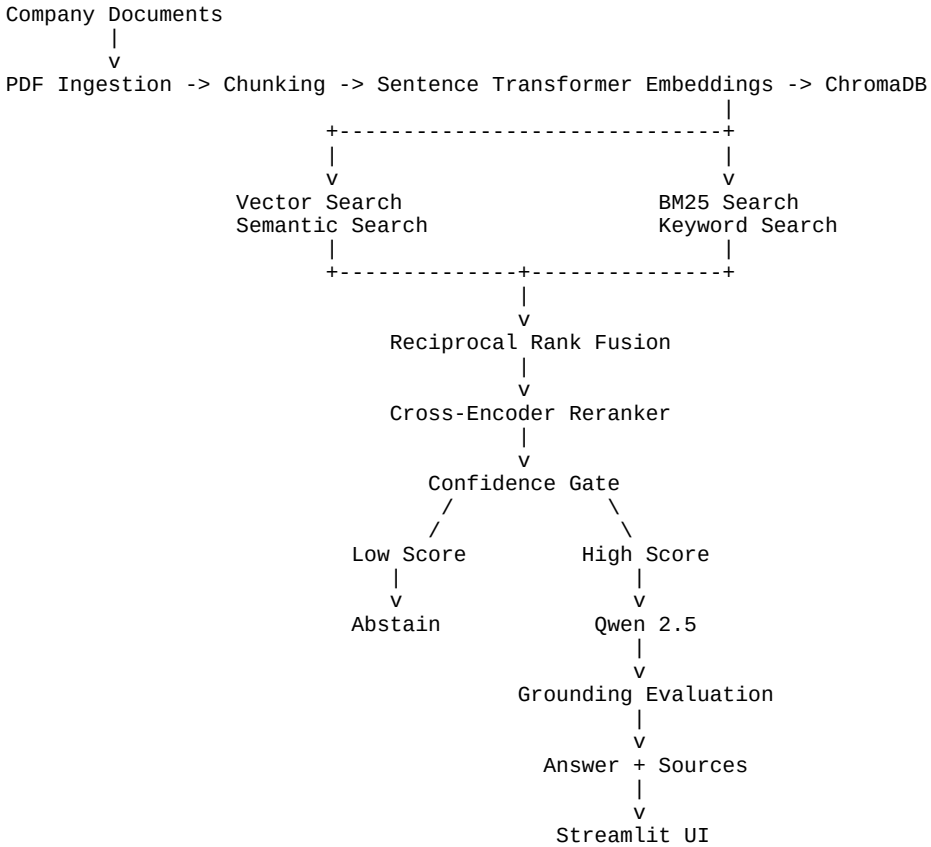
Project Overview

The Enterprise Knowledge Assistant is a production-style Retrieval-Augmented Generation (RAG) system designed to answer questions from company policies and internal documents. It combines semantic search, keyword search, Reciprocal Rank Fusion (RRF), cross-encoder reranking, confidence-based abstention, local LLM generation, and grounding evaluation.

Features

PDF document ingestion • document chunking • Sentence Transformer embeddings • ChromaDB • semantic vector search • BM25 • Reciprocal Rank Fusion (RRF) • cross-encoder reranking • confidence-based abstention • Qwen 2.5 through Ollama • grounding evaluation • source document display • Streamlit chat interface • retrieval and RAG evaluation metrics

Architecture



How the System Works

1. Document Ingestion

PDF documents are loaded and parsed using pypdf.

2. Chunking

Large documents are divided into smaller chunks. Metadata can include source document, chunk index, and page/document information.

3. Embeddings

Each chunk is converted into a numerical vector using a Sentence Transformer embedding model.

4. Vector Search

The question is embedded and compared with stored embeddings to retrieve semantically similar content.

5. BM25 Search

BM25 provides keyword-based retrieval and complements semantic search.

6. Reciprocal Rank Fusion

Vector and BM25 results are combined into a unified ranking.

7. Cross-Encoder Reranking

Candidate documents are scored based on the relationship between the question and retrieved document.

8. Confidence Gate

The highest reranker score is compared against a configured threshold. Weak retrieval causes abstention.

9. Answer Generation

High-confidence context is passed to local Qwen 2.5 through Ollama. The model is instructed to use only retrieved company information.

10. Grounding Evaluation

The answer is evaluated against retrieved context: SUPPORTED, ABSTENTION, or NOT_SUPPORTED.

11. Final Response

Streamlit displays the answer, grounding status, and sources.

Technology Stack

Python 3.12 | pypdf | Sentence Transformers | ChromaDB | BM25 | Reciprocal Rank Fusion | Cross-Encoder | Qwen 2.5 | Ollama | Streamlit | Custom Python evaluation pipeline

Project Structure

```
enterprise-knowledge-assistant/
■■■ app.py
```

```
■■■ README.md
■■■ requirements.txt
■■■ docs/evaluation.md
■■■ src/
■   ■■■ embeddings/embedder.py
■   ■■■ evaluation/grounding.py
■   ■■■ generation/generator.py
■   ■■■ ingestion/chunker.py
■   ■■■ ingestion/loader.py
■   ■■■ retrieval/hybrid_retriever.py
■   ■■■ retrieval/reranker.py
■   ■■■ vectorstore/chroma_store.py
■   ■■■ ingest.py
■   ■■■ rag_pipeline.py
■■■ tests/
■   ■■■ test_final_evaluation.py
■   ■■■ test_grounding_evaluation.py
■   ■■■ test_mrr.py
■   ■■■ test_retrieval_evaluation.py
■■■ data/
    ■■■ documents/
    ■■■ chroma/
```

Installation & Setup

```
git clone https://github.com/sq6111/enterprise-knowledge-assistant.git
cd enterprise-knowledge-assistant
python3.12 -m venv .venv
source .venv/bin/activate
python -m pip install --upgrade pip
python -m pip install -r requirements.txt
```

Ollama Setup

```
ollama pull qwen2.5:3b
ollama list
```

Expected model: qwen2.5:3b. Ollama must be running before the application.

Build the Vector Database

PDFs are in data/documents/. Generated ChromaDB is in data/chroma/.

```
python src/ingest.py
```

data/chroma/ is excluded from Git because it is generated locally.

Run the Application

```
source .venv/bin/activate
python -m streamlit run app.py --server.fileWatcherType none
```

Application: http://localhost:8501

Example Questions

```
Can I work remotely?
What should I do if my laptop is lost?
How do I get technical support?
What should a new employee do on the first day?
What is the company's business travel policy?
```

Abstention Example

```
Question:
What is the employee stock option policy?

Response:
I couldn't find that information in the company knowledge base.
```

This prevents the LLM from inventing information that is not present in the company documents.

Evaluation

Dataset: 7 answerable questions + 2 unanswerable questions = 9 total test cases.

```
Answerable Question Accuracy: 100.00%
Abstention Accuracy: 100.00%
Grounded Answer Rate: 100.00%
Expected Source Retrieval: 100.00%
Recall@3: 100.00%
Vector Search MRR: 1.0000
Hybrid Search MRR: 0.7381
Hybrid + Reranker MRR: 0.9286
```

Retrieval Performance

Vector search MRR = 1.0000. Hybrid retrieval using vector search, BM25, and RRF achieved MRR = 0.7381. After cross-encoder reranking, MRR increased to 0.9286, substantially improving ranking quality on the evaluation dataset.

Grounding & Abstention

Grounded Answer Rate = 100%. All answerable questions produced answers classified as SUPPORTED. Abstention Accuracy = 100%; the system correctly abstained when requested information was unavailable.

Evaluation Command

```
python -m tests.test_final_evaluation
```

Evaluation Questions

1. Can I work remotely? → 02_remote_work_policy.pdf
2. Laptop lost → 03_information_security_policy.pdf
3. Paid time off → 04_leave_and_attendance_policy.pdf
4. Technical support → 05_it_support_and_acceptable_use.pdf
5. First day → 07_employee_onboarding_guide.pdf
6. Business travel → 06_business_travel_policy.pdf
7. Information security requirements → 03_information_security_policy.pdf
8. Unanswerable: maternity leave
9. Unanswerable: employee stock options

Git Hygiene

```
.venv/
data/chroma/
```

```
__pycache__/
.env
.DS_Store
```

GitHub Repository

https://github.com/sq6111/enterprise-knowledge-assistant

Future Improvements

Larger evaluation datasets • query rewriting • conversation memory • authentication and access control • document-level permissions • additional document formats • retrieval latency monitoring • production deployment • automated evaluation pipelines

Conclusion

The project demonstrates an end-to-end modern RAG system combining semantic retrieval, keyword retrieval, RRF, neural reranking, confidence-based abstention, local LLM generation, grounding evaluation, and a Streamlit interface. Current evaluation achieved 100% Recall@3, 100% grounded answer rate, and 100% abstention accuracy; cross-encoder reranking improved hybrid MRR from 0.7381 to 0.9286.

Quick Terminal Setup

```
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python3.12 -m venv .venv
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python -m pip install --upgrade pip
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